

The relationship between a non-Western migration background and education level

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Set-up your environment

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Tutorial group 2-1

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

A study by research agency SCALIQ involving 6409 secondary school students, suggests that students with Arabic first and/or last names are more likely to be placed at a lower educational level than equally intelligent student with European sounding names. About half of the students with Arabic names were placed half a school level lower. The researchers say that this may be a sign of inequality in education. This is a problem because it can hurt students' education, career opportunities and motivation. However, the study only looks at intelligence and names. It did not look at other factors such as family income, home support or cultural background. The experts say that being placed in a lower school level that is too low can hurt students' education, confidence/motivation and career opportunities.

In another study by KIS they researched whether students with a migration background in the Netherlands receive a lower educational recommendation than they deserve for their abilities. Based on quantitative analyses and interviews with teachers, students and parents, the researchers found just a little evidence of discrimination based on ethnicity. However, factors such as Dutch language proficiency, home environment, classroom behavior and of course the perceptions of the teachers can influence the recommendations. This may negatively impact children with a different ethnic background.

These studies demonstrate that, to some extent, non-Western migrants in the Netherlands seem to obtain a lower level of education than the Dutch. While causal relationships are harder to determine, this gap in education is a social problem as it fuels income inequality, inequality in general, and disadvantages people with a migration background. The current politics and the widespread attention in the Netherlands for migrants, especially non-Western migrants, make this a relevant problem.

Part 2 - Data Sourcing

2.1 Load in the data

2.2 Provide a short summary of the dataset(s)

Summary dataset on education level and migration background

In the first dataset, which contains data on different education levels and migration background, a group consisting of both men and women was researched. All participants were 15 years or older. The dataset contains observations from 2003 to 2020 and contains data about the migration background, age, and level of education of participants for each year. It was sourced from the CBS (2021).

```
##          ID          Geslacht      Leeftijd      Migratieachtergrond
## Min.      : 190      Length  :12474      Min.      :20150      Length  :12474
## 1st Qu.: 41410      N.unique :    1      1st Qu.:52052      N.unique :   11
## Median : 83003      N.blank  :    0      Median :53500      N.blank  :    0
## Mean     : 82923      Min.nchar:    7      Mean     :46186      Min.nchar:    7
## 3rd Qu.:124316      Max.nchar:    7      3rd Qu.:53800      Max.nchar:    7
## Max.      :165536                                Max.      :53925
## HoogstBehaaldOnderwijsniveau      Perioden      Bevolking_1
## Min.      :2018710                                Length  :12474      Min.      : 0.0
## 1st Qu.:2018720                                N.unique :   18      1st Qu.:  2.0
## Median :2018770                                N.blank  :    0      Median : 16.0
## Mean     :2018764                                Min.nchar:    8      Mean     :130.7
## 3rd Qu.:2018800                                Max.nchar:    8      3rd Qu.: 77.0
## Max.      :2018810                                Max.      :3469.0
```

Summary dataset on migration background percentage per region

This dataset contains information of the percentage distribution of people with different backgrounds living in each province for 2026. It was sourced from AlleCijfers (2026).

```
##      Provincienaam % Herkomst Nederland % Herkomst buiten Nederland
## Length :12      Min.      :0.6100      Min.      :0.1300
## N.unique :12      1st Qu.:0.6950      1st Qu.:0.1900
## N.blank  : 0      Median :0.7650      Median :0.2350
## Min.nchar: 7      Mean     :0.7475      Mean     :0.2525
## Max.nchar:13      3rd Qu.:0.8100      3rd Qu.:0.3050
##                                Max.      :0.8700      Max.      :0.3900
## % Herkomst Europa % Herkomst buiten Europa
## Min.      :0.05000      Min.      :0.0800
## 1st Qu.:0.06750      1st Qu.:0.1150
## Median :0.08500      Median :0.1300
## Mean     :0.08917      Mean     :0.1633
## 3rd Qu.:0.11250      3rd Qu.:0.2175
## Max.      :0.14000      Max.      :0.3000
## % Geboren in Nederland met herkomst Nederland
## Min.      :0.6100
## 1st Qu.:0.6950
## Median :0.7650
## Mean     :0.7475
```

```

## 3rd Qu.:0.8100
## Max. :0.8700
## % Geboren in Nederland met herkomst Europa
## Min. :0.0200
## 1st Qu.:0.0300
## Median :0.0300
## Mean :0.0325
## 3rd Qu.:0.0325
## Max. :0.0600
## % Geboren in Nederland met herkomst buiten Nederland
## Min. :0.050
## 1st Qu.:0.080
## Median :0.095
## Mean :0.105
## 3rd Qu.:0.135
## Max. :0.170
## % Geboren in Nederland met herkomst buiten Europa % Geboren in Nederland
## Min. :0.0300 Min. :0.7700
## 1st Qu.:0.0475 1st Qu.:0.8325
## Median :0.0550 Median :0.8500
## Mean :0.0700 Mean :0.8517
## 3rd Qu.:0.1025 3rd Qu.:0.8900
## Max. :0.1400 Max. :0.9200
## % Geboren buiten Nederland met herkomst Europa
## Min. :0.03000
## 1st Qu.:0.04000
## Median :0.05500
## Mean :0.05583
## 3rd Qu.:0.07250
## Max. :0.09000
## % Geboren buiten Nederland met herkomst buiten Europa
## Min. :0.05000
## 1st Qu.:0.06750
## Median :0.07500
## Mean :0.09167
## 3rd Qu.:0.12000
## Max. :0.16000
## % Geboren buiten Nederland
## Min. :0.0800
## 1st Qu.:0.1100
## Median :0.1500
## Mean :0.1483
## 3rd Qu.:0.1675
## Max. :0.2300

```

Summary dataset on education level per region

This dataset contains information on the participation by people in different levels of education, expressed as a percentage, for each province. It contains data for the years 2023, 2024, and 2025. It was sourced from CBS (2025).

```

##      Regio's      Perioden
## Length :40    Min. :2023

```

```

## N.unique :14    1st Qu.:2023
## N.blank  : 0    Median :2024
## Min.nchar: 9    Mean   :2024
## Max.nchar:18    3rd Qu.:2025
##                               Max.   :2025
##                               NAs     :1
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Basisonderwijs, vmbo, mbo1 (%)
## Min.      :22.30
## 1st Qu.   :25.93
## Median    :26.55
## Mean      :26.18
## 3rd Qu.   :27.27
## Max.      :28.80
## NAs       :14
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Havo, vwo, mbo2-4 (%)
## Min.      :35.50
## 1st Qu.   :40.83
## Median    :42.70
## Mean      :42.51
## 3rd Qu.   :44.92
## Max.      :46.80
## NAs       :14
## Onderwijs|Naar woonregio|Hoogstbehaald onderwijsniveau|Hbo, wo (%)
## Min.      :24.90
## 1st Qu.   :27.75
## Median    :31.75
## Mean      :31.32
## 3rd Qu.   :33.10
## Max.      :42.20
## NAs       :14

```

2.3 Describe the type of variables included

The first dataset mentioned, `edu_mig_dataset`, contains demographic and SES information. The demographic information consists of age, sex, and background and the socioeconomic status information consists of education level. The two central variables in this dataset are background and education level, however, the additional information provides possibilities for more thorough analysis. The information has been obtained through the use of surveys (CBS, 2022). The dataset largely consists of characters, with the exception of the `Bevolking_1` column, which gives a numerical value for each row representing the amount of people that had the same results. Age, which can be a numerical value, has become an ordinal character as ages are grouped for every ten years. Education level is ordinal as well, but all other characters remain nominal.

The other two datasets, `reg_mig_dataset` and `reg_edu_dataset`, are very similar in format and variables. Both contain regional information for each of their variables, but they differ in the type of information. The first dataset contains demographic information, numerical values representing which percentage of the population has which background for each province. The second dataset contains SES information, also numerical values representing percentages, only now they represent the distribution of the population over different education levels for each province. The dataset on education has a column for each level of education. Both datasets have been sourced with open source data and/or administrative sources (CBS, 2025; AlleCijfers, 2026).

Part 3 - Quantifying

3.1 Data cleaning

Renaming unclear characters in edu_mig_dataset

The size of the dataset on education level and different migration backgrounds was so significant that all character values were replaced with codes. In order to have a meaningful dataset, all these codes were replaced with the character value the cells previously contained. This was done with the use of the legend provided by CBS.

Eliminating unnecessary observations and variables in edu_mig_dataset

Furthermore, the same dataset contained various variables and observations that were irrelevant to the analysis, like migration background data for singular countries and a column for sex, which only contained “total women and men” for all observations. By removing these unnecessary elements, the dataset is easier to use and has become one-fourth of its original size.

Renaming inconsistent characters in reg_edu_dataset

The dataset containing data on education levels for each province is to be used to compare regional education levels with the regional migration levels (reg_mig_dataset). However, these two datasets have used different names to address the same provinces, which can cause complications. Therefore, the names in the reg_edu_dataset have been replaced with those of their counterparts in the reg_mig_dataset to ensure uniformity.

Eliminating unnecessary years in reg_edu_dataset

The second change made to this dataset is the elimination of observations corresponding to the years 2023 and 2025. 2023 was eliminated since this dataset, as previously mentioned, will later be used for comparison and the dataset it will be compared to only contains the data for one year. 2025 was eliminated because it contained solely NA's, which made comparison impossible.

Eliminating unnecessary variables in reg_mig_dataset

The dataset on the proportion of people with a migration background per province contained numerous variables that were irrelevant for the intended use of the dataset. For example, the proportion of European migrants and a distinction between first and second generation migrants. These were meaningful data, however, they did not fit the purpose of this dataset.

3.2 Generate necessary variables

Variable 1: Percentage of the total of Dutch people and non-Western migrants in each level of education

This variable was taken from the cleaned version of the edu_mig_dataset. For the purposes of comparison, this variable was created to demonstrate how Dutch people and non-Western migrants are distributed across different levels of education and how this changes over time. In order to avoid bias because of group size, since there are more Dutch people than migrants, the total of Dutch people for each year was calculated and then the percentage proportion of Dutch people in each education level was calculated. The same was done

for non-Western migrants. This way it became clear how each group is distributed across the different levels of education. This resulted in the dataset `new_var_1`.

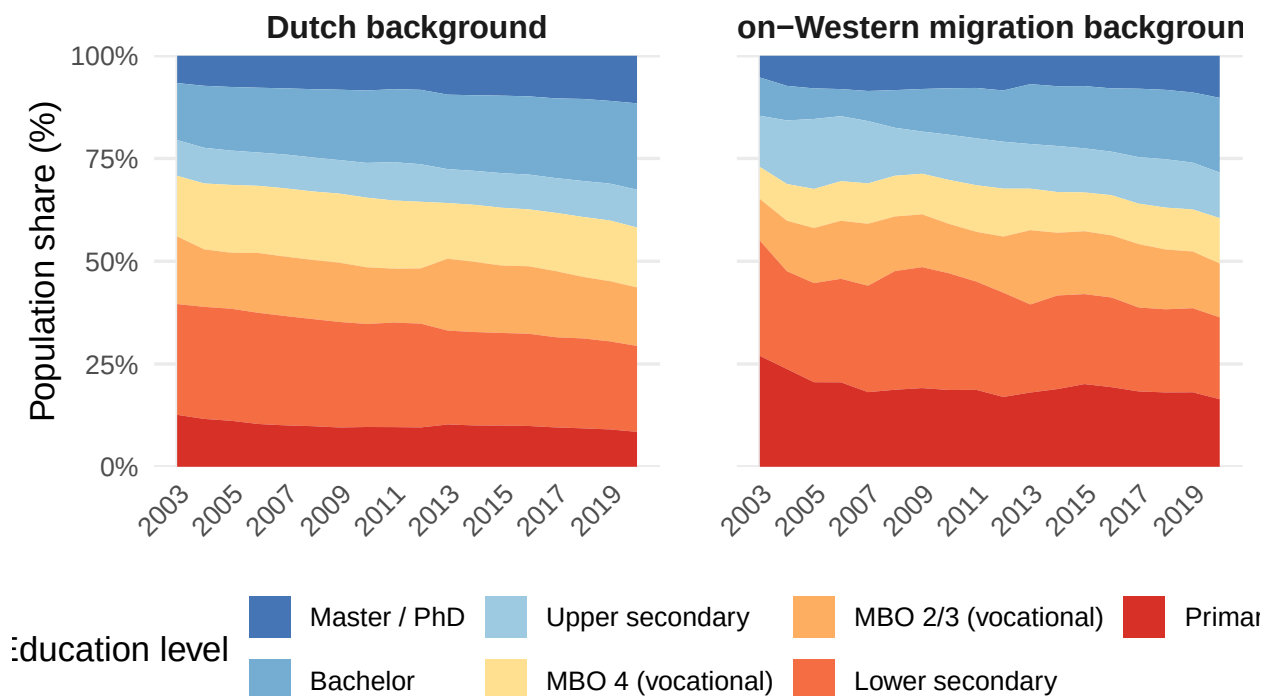
Variable 2: Percentage of people that have obtained a hbo- or wo-bachelor for different age groups in 2020

The second variable has also been created with data from the cleaned `edu_mig_dataset`. This variable was created with the purpose of visualizing sub-population variation, specifically the difference between age groups. The most recent year has been picked to create this variable, which was 2020. A hbo- or wo-bachelor has been selected because it qualifies as a high education. The data was grouped by age and migration background, excluding 15 to 25 year olds, since they would give a distorted image, as a part of this group is too young to obtain this level of education. Then a total of people per age group was computed, which was then used to calculate the percentage of their age group that obtained a hbo- or wo-bachelor. This resulted in a dataset called `combined_var`.

3.3 Visualize temporal variation

Highest level of education based on migration background

Population 15 year of older, 2003–2020 | source: CBS (EBB, tabel 8227



In the two plots above is shown how the education levels of the groups have changed over time. The left plot shows how the education level of the people with a Dutch background changed over time, and the right plot shows the same for people with a non-Western migration background. For the creation of these plots the first variable created earlier was used.

The plot for the people with a Dutch background shows that over time a bigger share of the population obtained a degree from a university or HBO. This is for mostly the bachelor degrees, but also for the master-

and doctor degrees there is an increase of the population share. The biggest decrease is seen for the group of people that have a lower secondary education or Vmbo-g/t.

The plot for the people with a non-Western migration background shows that over time a much bigger share of the population got a higher degree than in previous years. The biggest increase is seen in the group of people that have a bachelor degree. There is also a big decrease in the group of people that only finished primary education.

These plots show that the education level of people with a non-Western migration background is becoming closer to the the education level of people with a Dutch background.

3.4 Visualize spatial variation

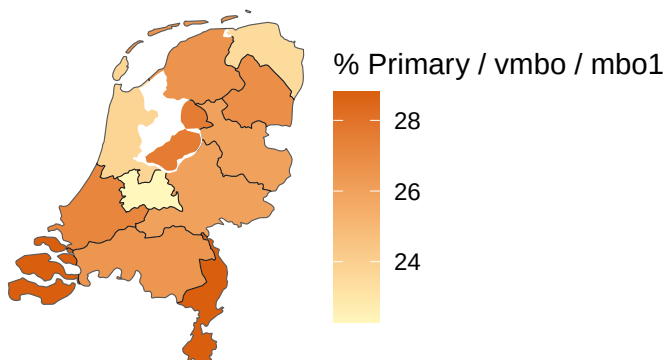
```
## Reading 'ne_10m_lakes.zip' from naturalearth...

## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries

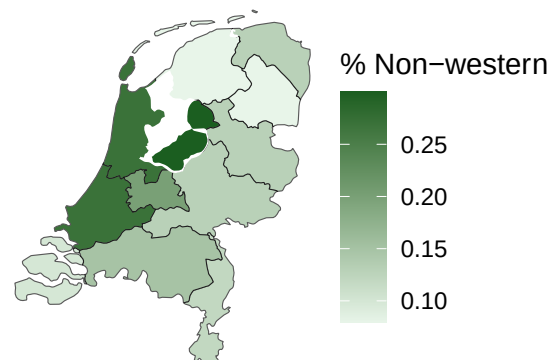
## Spherical geometry (s2) switched off
## although coordinates are longitude/latitude, st_intersection assumes that they
## are planar

## Warning: attribute variables are assumed to be spatially constant throughout
## all geometries
```

**Primary education or lower
per province**



**Non-Western migration background
per province**

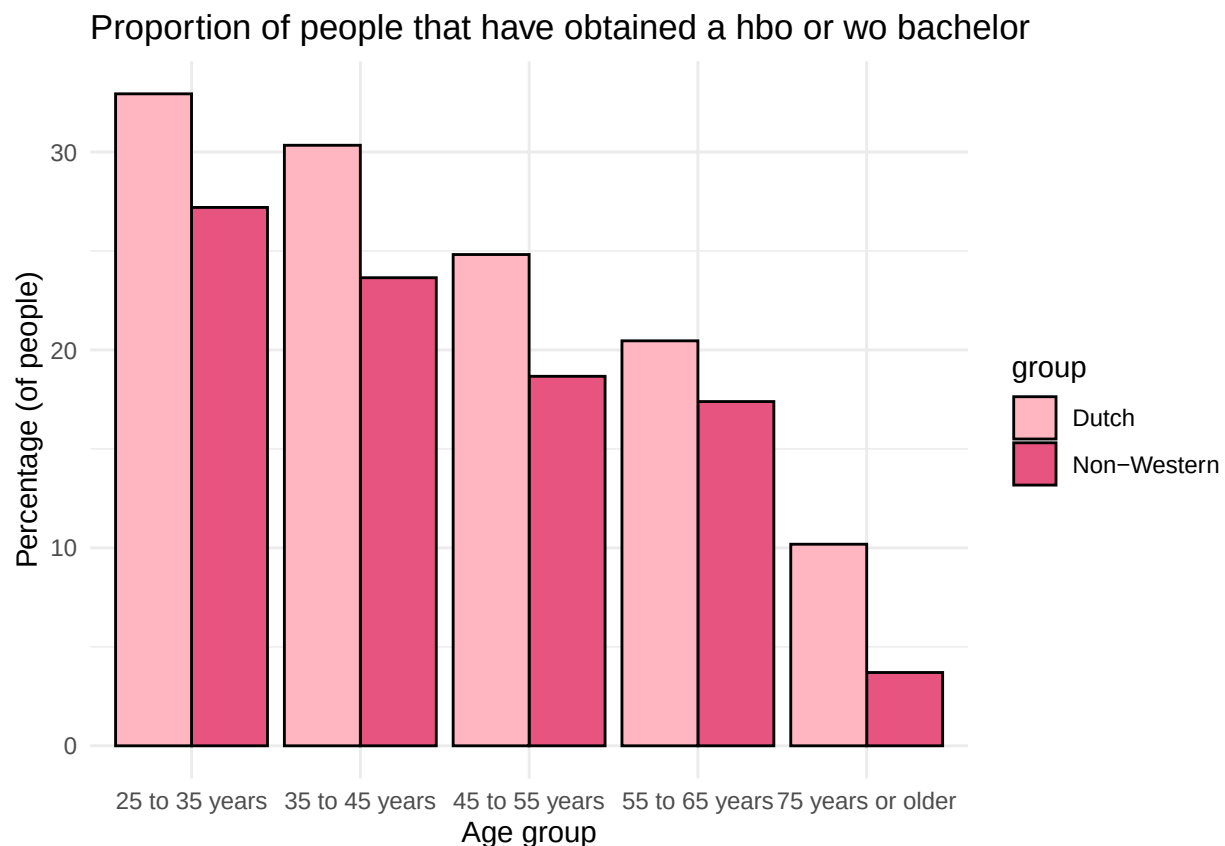


The left map shows the percentage of the population to have either only received primary education, vmbo-g/t or lower secondary education. The map shows that specifically in North-Holland, Utrecht and Groningen the education level is higher than in other parts of the country.

The map on the right shows the percentage of the population who have a non-Western migration background. The map shows that the percentage is the highest in the provinces that are a part of the Randstad. This shows that migration is mostly centered around these parts of the country. The biggest cities of the country are also in the provinces that have the highest percentage.

Overall, comparing the the graphs does not result in a clear resemblance, indicating that the proportion of non-Western migrants living in a region does not correlate with the proportion of low schooled individuals in that same region. The primary reason for these results seems to be that both high education and migration tends to concentrate itself in bigger cities, which is demonstrated in the graphs.

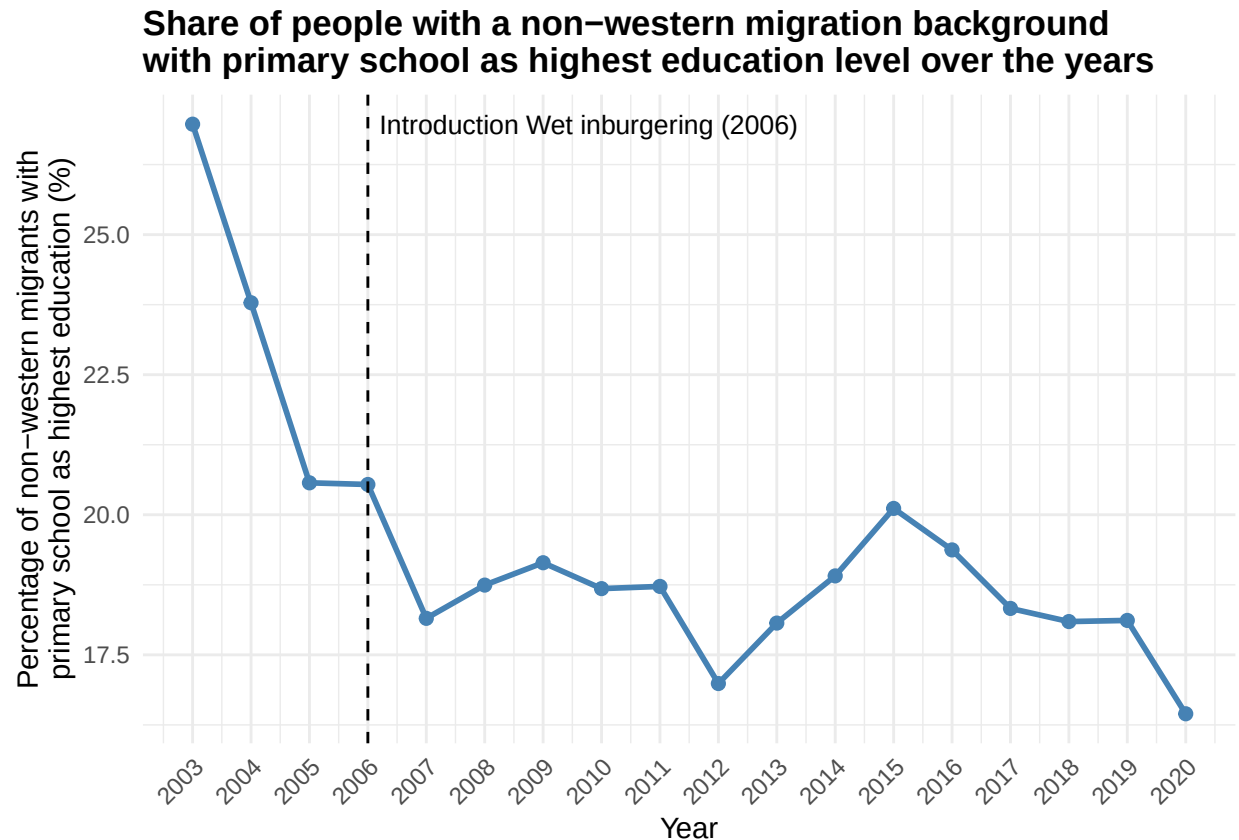
3.5 Visualize sub-population variation



The bar plot pictured above visualizes the percentage of people, both Dutch and with a migration background, that have obtained a hbo- or wo-bachelor in 2020 per age group. This is relevant to the social problem analysis, as it demonstrates not only the difference between Dutch people and non-Western migrants in their participation in higher education, but also whether this differs between different ages.

It is clear that, in general, the percentage of people hbo- or wo-bachelor has a negative correlation with age, demonstrating that the younger generations are becoming increasingly high educated. However, the proportional difference between Dutch people and non-Western migrants seems to stay approximately seven percent, except for the 55 to 65 years group. These findings show that the gap does not seem to be closing in the younger generations.

3.6 Event analysis



In the graph above the share of people with a non-Western migration background and with primary school as their highest education level is shown. On the 30th of November in 2006 there was a new law “Wet inburgering” and article 1d of this law was about compulsory school age (“Wet Van 30 November 2006, Houdende Regels Inzake Inburgering in De Nederlandse Samenleving (Wet Inburgering),” 2006). This law stated the compulsory school age for people who are staying in the Netherlands would be the same as that for Dutch citizens. This event is shown in the graph with the dashed line in 2006.

The graph shows that the share of people only having primary education declines significantly throughout the years. It is also shown that right after the introduction of this law there is a steep decline. Thus people with a non-Western migration background have a higher education level.

Part 4 - Discussion

4.1 Discuss your findings

The results of the analysis seem to indicate that there is a relationship between a non-Western migration background and the level of education the average person obtains. This relation is negative, as the presence of a non-Western migration background in general seems to correspond with lower average levels of education. The data show that non-Western migrants make up a much larger share of people with only primary or lower level education, while most Dutch people move beyond these levels of education. The temporal visualization does demonstrate that this educational gap is becoming smaller. In 2003, primary education was the highest education for 27% of migrants and 13% of Dutch people. In 2020, these percentages shifted to 16% and

8% respectively, decreasing the gap with 42.9%. The gaps between other levels of education have moved similarly in this time frame, demonstrating that the problem is shifting.

Geographically, the social problem does not seem to be location bound. When comparing a graph in which the proportion of people that have a low education per province with the proportion of non-Western migrants per province, the results do not seem to overlap, while other results indicate that this relation is existent. The primary explanation is, as mentioned before, that both migrants and highly educated people seem to concentrate around big cities, as they possess the most housing and career opportunities. Moreover, this visualization cannot be used to produce a correct relationship on its own, as many more variables need to be considered. It is merely a graphical comparison to evaluate whether the relation as described, affects the geographical distribution as well.

Furthermore, the sub-population visualization demonstrated a trend of younger people becoming increasingly educated, with the percentage of non-Western migrants and Dutch people that have obtained a hbo, or wo bachelor being 3.7% and 10.2% for ages 75 and older, while being 27.2% and 32.9% for ages 25 to 35. The gap in between backgrounds, however, seems to stay in between five to seven percent. There seems to be a small decrease in the gap in between generations, approximately 0.8%, but whether this indicates a trend of progress is difficult to conclude.

The trend for non-Western migrants with only primary education as the highest achieved level, shows an initial decline in 2006 after the introduction of the Wet inburgering policy. The trend then levels off at around 18%-20%, which means that this policy only helped a little.

In conclusion, analysis shows that people with a non-Western migration background on average obtain a lower education in comparison to a native Dutch person. However, this educational gap is evidently decreasing over time. The sources of this inequality could be many as discussed in Part 1, for example a language barrier, a lack of education in the country of origin, or discrimination in the system. The results show that these problems can be solved, as the gap is decreasing, posing a possibility for the government to aid and speed up the process of equality in schooling with initiatives like extra Dutch classes to overcome the language barrier and extra defenses against discrimination.

4.2 Limitations of results

The analysis that resulted in these results was limited, so some limitations need to be addressed. First and foremost, the relationship that these results demonstrate is not proven to be causal and not intended to be interpreted that way. It is very likely that a confounding variable is present, like the sources of the educational gap as mentioned before, that lead to this gap, or create a large part of it.

Another limitation is the source of one of the datasets, the dataset on migration background and education (`edu_mig_dataset`). This dataset has been constructed by the CBS through the use of the working population survey. This survey excludes all people in institutions or care homes, which excludes refugee centers (CBS, n.d.). This leads to a slightly distorted image as some non-Western migrants are not represented in the data.

Part 5 Reproducibility

A `renv` lockfile was made to ensure reproducibility of the entire project. It can be found within the GitHub repository link below.

5.1 Github repository link

<https://github.com/RikdeBruijn/PFE2026>

5.2 Reference list

AlleCijfers (2026, June 8). “Ranglijst: autochtoon en migratieachtergrond van de bewoners per provincie in Nederland”. Retrieved June 8, 2026, from: https://allecijfers.nl/ranglijst/autochtoon-en-migratieachtergrond-per-provincie-in-nederland/#google_vignette

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Kind met Arabische naam krijgt vaak lager schooladvies dan even slim kind met Nederlandse naam. (2025, februari). NU.NL. Geraadpleegd op 17 juni 2026, van <https://www.nu.nl/binnenland/6347333/kind-met-arabische-naam-krijgt-vaak-lager-schooladvies-dan-even-slim-kind-met-nederlandse-naam.html>

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